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Affecting voters through eyes and mind? An eye-tracking study to understand how political preferences affect attention and memory of political advertisements

Puck C. Guldemond, Sophie C. Boerman, Susan A. M. Vermeer, and Sanne Kruikemeier

ABSTRACT
While online political targeting is increasingly used in political campaigns, its effect on voters' attention toward political advertisements and the extent to which they are remembered remain unclear. This paper reports a within-subjects eye-tracking study ($N = 135$) to determine whether voters spend more time looking at political advertisements aligning with their political preferences while performing a goal-directed task. Moreover, it examines whether voters recall and recognize political advertisements that align with their preferences better. The results indicate mixed effectiveness, challenging the assumption that political advertisements from a preferred party attract greater visual attention and deeper cognitive processing. Overall, the notion that targeted political ads inherently capture voters' attention due to higher self-relevance or political alignment is not supported. This study contributes to understanding the complexity and nuances of targeted political advertising's effectiveness in engaging voters' attention and cognitive processes.

KEYWORDS
Eye-tracking; selective exposure; cognitive responses; political communication; information processing; political advertising

Political targeting has become a widely employed strategy in modern political campaigns to engage voters, yet it raises significant concerns regarding its impact on democratic processes. Political (micro)targeting is generally understood as a data-driven technique that uses digital traces people leave online to generate insights into the campaign's target audience (Dommett et al., 2023). The public debate about it has had a critical tone, in which governments and media pundits have expressed their worries about its possible impact on democracy. While some argue that political targeting forms a threat to society by manipulating or suppressing voters (Zuiderveen Borgesius et al., 2018), others believe it has limited effects (Baldwin-Philippi, 2019). As a consequence, scholarly consensus about the impact of online political targeting remains inconclusive. Nevertheless, it is often assumed that targeted political advertisements have a persuasive advantage because they are tailored to align with the preferences of individual recipients.

Findings from the literature on personalized advertising support the assumption that targeted political advertisements possess a persuasive advantage. Research indicates that personalization increases attention, elaboration, processing, and ultimately persuasion due to the higher perceived self-relevance of the advertisement (Boerman et al., 2017; De Keyzer et al., 2015; Teeny et al., 2021). In essence, voters are more responsive to personalized advertisements because they perceive them as more personally relevant compared to generic ads. Furthermore, evidence indicates that personalized advertisements capture greater visual attention, driven by an attentional bias toward self-relevant stimuli (Köster et al., 2015).

However, political (micro)targeting is a moving field characterized by conceptual and empirical ambiguity (Baldwin-Philippi, 2017, 2019; Bennett & Gordon, 2021). As Bennett and Gordon (2021) argue, political (micro)targeting represents a continuum that spans from highly individualized voter communication to broader, demographically defined messaging. This definitional fluidity complicates assumptions about when and why targeted political advertisements should yield persuasive advantages. Drawing on insights from personalized advertising, the political targeting literature often posits that advertisements tailored to recipients' preferences, such as their political preference or

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interest in the depicted issue, should attract greater attention and have stronger effects. Yet, it remains unclear whether such attentional advantages consistently translate to the political context.

Prior research suggests that political advertisements aligned with voters' predispositions capture more visual attention. For example, studies by Marquart et al. (2016) and Schmuck et al. (2020) both show that political preference significantly drives visual attention toward political posters. More specifically, both studies find that people allocate more visual attention to political posters that align with their political preferences. These findings align with the broader literature on personalized advertising, which shows that advertisements perceived as personally relevant, such as those promoting a preferred political party, capture more attention (Bang & Wojdynski, 2016; Bang et al., 2019; Köster et al., 2015).

However, the experimental design of previous studies (Marquart et al. 2016; Schmuck et al. 2020) involved presenting political advertisements side by side. We argue that this setup is less externally valid in an online environment: it does not accurately mimic patterns of exposure because advertisements are rarely presented as such. Rather, advertisements in online contexts typically appear when people read news online or scroll through their social media feeds, such as Instagram or Facebook. To address these limitations and advance our understanding of political ad effectiveness in more ecologically valid online environments, this paper presents a novel within-subjects eye-tracking study. We examine whether visual attention, recall, and recognition are greater for political advertisements that align with individuals' political preferences compared to those that do not. Additionally, we investigate the extent to which political ads capture participants' attention during a goal-directed task, specifically, reading online news.

Our contributions are twofold: firstly, most research on targeted political advertising has been conducted in the United States, which has a two-party system. Countries with multi-party systems, such as the Netherlands, offer a vastly different, more fragmented political landscape (Harteveld, 2021). With many parties competing for attention, Dutch voters are exposed to a broader range of targeted political advertisements. Yet, empirical insights into how these ads influence voters remain limited. Secondly, by employing eye-tracking, we can examine whether the presumed persuasive advantage of targeted political advertisements manifests in the earliest stage of message processing, namely, visual attention. By doing so, we gain new, nuanced insights into what drives attention to targeted political advertisements and how it influences cognitive processing. Thirdly, eye-tracking enables us to move beyond self-reports in measuring exposure and attention to political advertisements (Vraga et al., 2016), allowing for a precise assessment of visual attention and the degree to which such ads draw focus.

Do political advertisements that match individuals' political preferences attract attention?

Despite the often-assumed persuasive advantages of political targeting, we still know little about whether political targeting enhances visual attention, recall, and recognition of political advertisements. When interacting in today's high-choice media environment, people are exposed to a mixture of content such as posts, news articles, and advertisements. As people's attention and processing capacity are limited (Lang, 2000), there is stark competition within online environments to attract users' attention (Marquart, 2023). In such competitive media environments, information overload is likely to occur. To reduce the complexity of high-choice media environments, individuals employ cognitive mechanisms that enable them to select and focus on particular input for further processing, while simultaneously disregarding irrelevant or distracting information (Kohout et al., 2023; Stevens & Bavelier, 2012). This process is called selective attention, which is embedded in the selective exposure paradigm. Selective exposure is defined as "any observable bias of exposure to available communication content, captured through behavioral measures of media use" (Knobloch-Westerwick, 2015, p. 960). In other words, people selectively attend to some messages while disregarding others, which also means that they process some messages deeper than others.

Previous studies show that people tend to avoid information that they consider to be

advertising, which is referred to as "banner blindness" (Benway, 1998). Additionally, people with low political interest tend to skip any online content that is related to politics (Bode et al., 2017), and individuals with higher levels of self-efficacy tend to avoid political advertisements more often (Minihold et al., 2025). If people skip over, avoid, or only briefly attend to (political) advertisements, their attention to it is very limited. Such behavior prevents them from deeply processing and being able to remember information about the substance of the (political) content they were exposed to (Lang, 2000).

To ensure that the advertised message is processed and remembered, advertisers must find a way to capture the audience's attention. Thus, advertisers tailor their content to the preferences of the audience to make the advertisement personally relevant to them, which is better known as personalized advertising (Boerman et al., 2017; Li, 2016). Research in the commercial sector indicates that personalized advertising improves (visual) attention and message processing, among other benefits, by increasing the perceived personal relevance of an advertisement (Bang & Wojdynski, 2016; Boerman et al., 2017; Köster et al., 2015; Yeo et al., 2025). When an advertisement is perceived as personalized, it is congruent with one's self-concept, which increases the perceived relevance of the advertisement (De Keyzer et al., 2022; Yeo et al., 2025). In other words, incorporating personal information in advertisements increases the feeling that the advertisement was designed specifically for the person receiving it. When personalized advertisements draw more visual attention, people spend more time looking at them, which also means that they have more time to process them. In turn, this leads to a higher memory (i.e., recall and recognition) of the advertisement (Köster et al., 2015). For political advertising, higher perceived personal relevance can be achieved by creating a fit between the targeted political ad and the political preferences of the message receiver (Binder et al., 2022; Hirsch et al., 2023). This can be done in two ways: targeting people on their preferred political party, which they intend to vote for, or targeting people on their stances regarding political issues.

Targeting based on individuals' preferred political party

For political parties, the electorate can be roughly divided into three groups: their electoral base, swing voters, and the base of other political parties (Endres & Kelly, 2018). By dividing the electorate into subgroups, political parties aim to optimize their campaigns by mobilizing their electoral base, persuading swing voters, and avoiding wasting resources on the electoral base of other parties (Dommett et al., 2023; Endres & Kelly, 2018; Lavigne, 2021). Political targeting allows them to zoom in on these different subgroups and send them tailored messages based on the audience's preferences. Previous studies found that political targeting can effectively reinforce party ties (Lavigne, 2021), that political parties can benefit from targeting within their electoral base (Noetzel et al., 2024), and that receiving more advertisements from a particular party positively influences voters' preference for that party (Chu et al., 2024). Furthermore, Vliegenthart et al. (2024) show that targeted political ads from a liked political party are considered more acceptable than those from a non-liked party.

These positive effects can be attributed to how voters process targeted political advertisements. Specifically, congruent information is thought to elicit greater cognitive involvement, facilitating better information processing and improving the storage and retrieval of information (Lang, 2006). Moreover, individuals tend to process information with a confirmation bias, favoring content that aligns with their interests and predispositions (Erisen et al., 2014; Taber & Lodge, 2006). Consequently, they are more likely to expose themselves to advertisements consistent with their preferences.

Empirical evidence supports this tendency: studies by Marquart et al. (2016) and Schmuck et al. (2020) indicate that individuals allocate more visual attention to political advertisements that align with their political ideology. These findings highlight the selective exposure mechanism in how

individuals engage with political advertisements. Increased visual attention, in turn, enhances the likelihood of people processing the content of the advertisement. The relationship between visual attention and information processing can be explained by the eye-mind assumption, which posits a direct link between where someone looks and what they cognitively attend to. As such, increased visual attention leads to more opportunities to process, store, and eventually retrieve messages (Just & Carpenter, 1980; King et al., 2019; Lang, 2000). Thus, allocating visual attention to political advertisements should coincide with higher ad memory (Lee & Ahn, 2012).

Taken together, people are more likely to allocate their visual attention to targeted political advertisements because they are tailored to their individual preferences. As people allocate their visual attention to targeted political advertisements, they are more likely to process the message, enabling them to store and retrieve the given information better. Based on the discussed literature, we hypothesize:

H1: The extent to which a political ad matches political preferences (i.e., voting intention for the depicted party) positively affects a) visual attention to the ad, and b) ad memory (i.e., ad recall and ad recognition).

Targeting based on individuals' stances regarding political issues

When targeting voters based on issues, political parties can choose to either focus on issues that are owned by themselves or to focus on issues owned by rival political parties (Hillygus & Shields, 2008; Thiele & Seliger, 2025). In essence, issue-based targeting focuses on political issues that voters strongly care about, which are sometimes described as wedge issues (i.e., divisive political issues that highlight political cleavages) (Dobber et al., 2023; Endres, 2020; Hillygus & Shields, 2008). Research conducted in the United States illustrates that wedge issues can pull away voters from other parties (Endres, 2020; Endres & Panagopoulos, 2019). For example, in a two-party system like the US, it could be that a Republican voter who believes that every woman should have the right to an abortion feels torn between the stance of the Republican party on abortion and their own opinion on the topic of abortion. This conflict can create cross-pressure, where voters experience tension between their own stance on an issue and the stance of their preferred political party. As a consequence, voters might be persuaded to vote differently or abstain from voting altogether (Endres & Panagopoulos, 2019). Thus, targeting voters based on (wedge) issues can be an effective strategy to pull away votes from an opposing political party.

Studies conducted outside the US, with different political systems, also emphasize that issue-based targeting can be a successful strategy. A study that collected data from 25 countries shows that people find targeted political advertisements more acceptable when the ads depict a political issue that was rated by the participants as important to them (Vliegenthart et al., 2024). Moreover, studies conducted in the Netherlands, a country with a multiparty system, found that political advertisements highlighting issues important to the recipient are perceived as more interesting and persuasive, and that exposure to online ads increases the salience of these issues among voters (Chu et al., 2023; Vrielink et al., 2025). Moreover, advertisements that align with voters' opinions have been found to enhance political participation (Matthes & Marquart, 2015). As Dobber et al. (2023) argue, issue-based targeting can appeal to voters' self-concept by making the political advertisement personally salient and by introducing a powerful argument based on the voters' personal situation, beliefs, and behavior.

Building on the literature mentioned earlier, we expect that voters will devote more visual attention to, and exhibit higher recall and recognition of, political advertisements that address issues that are important to them. This effect is driven by the alignment between the issue and their self-concept, which enhances the perceived relevance of the advertisement (Bang & Wojdynski, 2016; Boerman et al., 2017; De Keyzer et al., 2022; Köster et al., 2015; Teeny et al., 2021). Thus, we expect:

H2: The degree of topic importance of the topic depicted in a political ad increases a) visual attention to the ad and b) ad memory (i.e., ad recall and ad recognition).

Reinforcing effect of targeting on party preference and issue stance

When a political advertisement stems from a political party that the receiver is likely to vote for and thus matches the party preference of the message receiver and depicts a political issue they strongly care about, it may be perceived as even more personally relevant. More specifically, we expect that political advertisements aligning with both the party preference and issue position of the recipient will amplify the effects in such a way that people allocate even more visual attention to it and will be better able to remember the ad. However, in some cases, matching a message to the personal preferences of its receiver can backfire. For example, due to increased privacy concerns or weak arguments presented in the advertisement (Boerman et al., 2017; Teeny et al., 2021). Additionally, a recent study shows that targeting voters on several personal characteristics does not offer any persuasive advantage over targeting them based on a single data point (Tappin et al., 2023). Hence, based on the mixed findings in existing literature, we propose the following research question:

RQ: *To what extent do the effects of party preference and topic importance on a) visual attention to the ad, and b) ad memory (i.e., ad recall and ad recognition) interact?*

Method

Design and sample

We conducted a preregistered[1] within-subject eye-tracking study that consisted of a short survey before exposure to the stimuli, and again a survey after participants were exposed to the stimuli. The stimuli consisted of six news article leads accompanied by advertisements (see stimulus material in the online Supplementary materials for more details). All participants were exposed to the 6 advertisements, but the order of exposure to the stimuli was chosen randomly for each participant. The design of this study was approved by the Ethics Committee of the University under approval number 2023-050. Participants were recruited using the services of a Dutch research company. A priori power analysis with 1 - $\beta = .80$ and $\alpha = .05$ revealed that a sample of $N = 81$ is needed to detect small effects ($f^2 = .10$).[2] After filtering out participants who had difficulties during calibration, for example, because they had an eye condition or because data quality was low ($n = 11$), our final sample consisted of 135 participants, which is a large sample size for eye-tracking studies (King et al., 2019). The final sample consisted of 66 women (48.9%) and 69 men (51.1%), and had a mean age of 47.63 ($SD = 18.28$). The majority of our sample (65.9%) indicated having a higher education level, and 34.1% indicated having a lower education level. Meaning that our sample is approximately representative of the Dutch population. Finally, on a 7-point left-right scale ($M = 3.54$, $SD = 1.67$), 54.8% indicated their political preference to be left of the center.

Procedure

First, participants were welcomed at the University, where they were placed in the lab together with the responsible researcher. They filled in a pretest survey measuring our independent variables (see measures) and provided informed consent on an iPad. We included some filler and distraction questions to mask the true purpose of our study (e.g., which of the following influencers, celebrities, or accounts do you follow on Instagram?). After filling out the pretest, participants were placed behind a Tobii Pro Spectrum at approximately 65 cm distance from the computer screen. Participants were given a goal-directed task: they were instructed to read several leads of news articles as they would normally do. When they had finished reading the article, or if they wanted to proceed to the next article, they had to click once with the mouse. They were not able to go back to the previous article. Before reading the news articles, the eye-tracker was calibrated with a 9-point calibration procedure, which resulted in a mean score of 0.23 for precision ($SD = 0.21$) and a mean accuracy score of 0.50 ($SD = 0.21$). If the calibration results exceeded 1° for precision or accuracy (or both), the calibration process was repeated until data quality was sufficient. After reading the articles, participants took a posttest survey on the iPad, measuring our

dependent and control variables. Data was collected between 9 October and 2 November 2023. On 22 November 2023, the national elections were held in the Netherlands.

Stimulus material

We used the leads of six news articles accompanied by a static advertisement in the lower-right corner (see Appendix A). Every news article was always paired with the same advertisement. We used Tobii Pro Lab to create areas of interest (AOI). The AOIs were placed around the advertisements, and we made sure that they were the same size (see Appendix B). We chose static rather than dynamic stimuli in the lower-right corner because previous studies found that banner advertisements containing animation attract less attention and decrease ad memory (Lee & Ahn, 2012). Moreover, we placed the stimuli in the lower-right corner as previous studies indicate that the optimal position of a banner is on the right-hand side, and banners in the lower-right corner capture more attention compared to banners in the upper-right corner (Köster et al., 2015; Lim & Wogalter, 2000; Simola et al., 2011).

Two of the advertisements were nonpolitical, while the other four were mock political advertisements. Two political advertisements stemmed from the People's Party for Freedom and Democracy (*VVD*), and two from the Green Left-Labor Party (*GroenLinks-PvdA*). The VVD is characterized as a right-wing liberal political party, while GroenLinks-PvdA is a left-wing green social-democratic political party. At the time of the experiment, VVD was the biggest governing party, while GroenLinks-PvdA was the biggest opposition party.[3] Since effect sizes are small, we selected two ideologically distinct parties, as we expect visual attention to differ more between dissimilar than between similar parties. We used the house style of both political parties to create the political advertisements, which can be found on their websites, to make sure the design of the mock advertisements was close to real advertisements of both parties. Canva was used to design the advertisements, and we made sure that the design of the political advertisements was as similar as possible.

Of the right-wing advertisements, one depicted the party leader and the party logo of VVD, while the other depicted the topic of climate, accompanied by the party logo. Of the left-wing advertisements, one depicted the party leader and the party logo of GroenLinks-PvdA, while the other depicted the topic of safety and the party logo. This was the first general election for both party leaders as lead candidates. We chose to use pictures of the party leaders in a professional setting as previous studies show that such images are more attention-grabbing than those portraying politicians in private contexts (Lindholm et al., 2021). The topic climate is typically owned by GroenLinks-PvdA, while the topic safety is typically owned by VVD. The design of the issue-based advertisements allows us to test whether issue importance can draw visual attention across party lines, as the design intentionally exposes participants to issue-based advertisements from parties they do not prefer and that do not own the depicted issue. In other words, it allows us to examine whether left-wing voters who consider climate a very important topic pay more visual attention to the right-wing advertisements about climate change, even though these ads come from an opposing party.

Measures

Independent variables

The independent variable, political preference, was measured by asking, *"How likely is it that you would vote for the following parties if there were elections for the parliament in the Netherlands tomorrow?"* Participants had to indicate their vote intention for all 18 parties represented in the parliament before the Dutch 2023 general elections on an 11-point scale (1 = *very unlikely*, 11 = *very likely*). We measured vote intention rather than partisanship or general left – right ideology because vote intention provides a more precise indicator of political orientation in a multi-party system like the Netherlands. Among the 18 parties for which we measured participants' vote intention were the left-wing party GroenLinks-PvdA ($M = 5.54$, $SD = 3.96$) and the right-wing party VVD ($M = 5.08$, $SD = 3.77$), which were the parties of interest in this experiment. To improve interpretability, we refer to vote intention for the VVD as right-wing

vote intention and vote intention for GroenLinks-PvdA as left-wing vote intention, reflecting the widely recognized ideological positions of these parties in the Dutch political landscape. The independent variables topic importance of safety (M = 5.99, SD = 1.02) and climate (M = 5.94, SD = 1.11) were measured by asking participants: *"Could you please indicate how important the following topics are to you?"* on a 7-point scale (1 = *not important at all*, 7 = *very important*).[4] The topics of safety and climate were included in a list of eight topics in total to avoid priming participants on the topics of interest for this study. The other topics consisted of education, infrastructure, housing market, immigration, economy, and defense.

Dependent variables

We operationalize visual attention with the eye-tracking metric total fixation duration. This metric represents a sum of the duration of all fixations that were identified per participant in the Area of Interest (AOI) specified by the researcher and is often used as an indicator of cognitive processing (King et al., 2019). Fixation-based metrics are the most commonly reported metric in eye-tracking studies within advertising and (political) communication research to capture how long people fixate on a specific AOI (King et al., 2019; Segijn, 2025). Visual attention was measured with a Tobii Pro Spectrum, which has a sampling rate of 500 Hertz. This eye-tracking device operates unobtrusively as it is not attached to participants in any way. The eye-tracker was installed under a 23.8-inch computer screen, on which the stimulus material was presented. This creates a natural environment to measure eye movement and enables us to measure fixation time (in seconds) toward specific AOIs (see Table 1 for an overview of descriptive statistics), which were placed around the advertisements (see Appendix B).

Recall was measured by asking participants: *"Did you see any advertisements while you were reading the news articles on NU.nl?"* In total, 83.7% of the participants indicated that they had seen advertisements, 7.4% indicated they had not seen any, and 8.9% indicated that they could not remember whether they had seen any. The 83.7% of participants who indicated they indeed saw advertisements while reading the news articles were asked the following question: *"Can you remember who or what the senders of these ads were?"* Participants were instructed to fill in as many ad senders as they could remember, without taking spelling into account. The VVD was mentioned by 63.7% of the participants, and GroenLinks-PvdA was mentioned by 65.2% of the participants.

After recall, recognition was measured by asking people to indicate which advertisements they had seen while they were reading the news articles. Participants were shown 9 advertisements, of which 3 were included as filler ads, meaning that they were not displayed as stimulus material during the experiment. The political ad of the VVD depicting the party leader (see Appendix A1.1) was recognized by 60.7% of the sample. The political ad of GroenLinks-PvdA depicting the party leader (see Appendix A1.2) was recognized by 71.1% of the participants. The political ad of the VVD depicting the issue of climate (Appendix A1.3) was recognized by 45.9% of the participants. And lastly, the political ad of GroenLinks-PvdA depicting the issue of safety (Appendix A1.4) was recognized by 46.7% of the sample.

Control variables

Age, gender, and level of education were added as control variables (see design and sample for the descriptive statistics). In addition, we controlled for the time participants spent reading the news articles, the order in which the stimuli

Table 1. Descriptive statistics of total fixation duration in seconds.

	Range (min – max)	*M*	*SD*	Participants who did not fixate on the advertisements*
Political ad: right-wing + party leader	0 – 6.19	1.19	1.33	19.3
Political ad: right-wing + issue climate	0 – 7.51	1.03	1.32	23.7
Political ad: left-wing + party leader	0 – 4.61	0.90	0.82	9.6
Political ad: left-wing + issue safety	0 – 9.19	1.20	1.61	25.2
Nonpolitical ad: Starbucks	0 – 6.50	0.98	1.23	28.9
Nonpolitical ad: Kwantum	0 – 4.23	0.64	0.78	25.9

*Scores represent the percentage of participants with a fixation time of 0.0 seconds.

were presented, the number of days until the Dutch elections, and participants' political interest. As political interest might steer attention toward political advertisements, we asked participants to what extent they agree with the following statement: *"I am generally interested in politics"* (1 = *totally disagree*, 7 = *totally agree*; *M* = 5.19, *SD* = 1.42). We used these control variables because the more time people spend reading the articles, the more opportunity they have to look at the accompanying (political) advertisements. Moreover, we believe it is important to control for the order in which the stimuli were presented because when people notice that the news article is accompanied by advertisements, they are likely to notice the advertisements that follow too. Some people may have noticed the advertisements directly, while others may have noticed them later during the study.

Results

Descriptive statistics

Table 1 provides an overview of the minimum and maximum time participants spend looking at the AOI that was placed around the advertisements, as well as the mean and standard deviation. In addition, Figure 1 shows the distribution of total fixation time per ad. Notably, it shows a peak around 0, indicating that the majority of participants did not fixate or fixated only briefly on the political advertisements. More specifically, between 9.6% and 25.2% of participants did not fixate on the political advertisements, and between 54.8% and 81.5% of participants only fixated between 0 and 2 seconds on the political advertisements. Appendix C provides heatmaps of the stimulus displaying visual attention distribution.

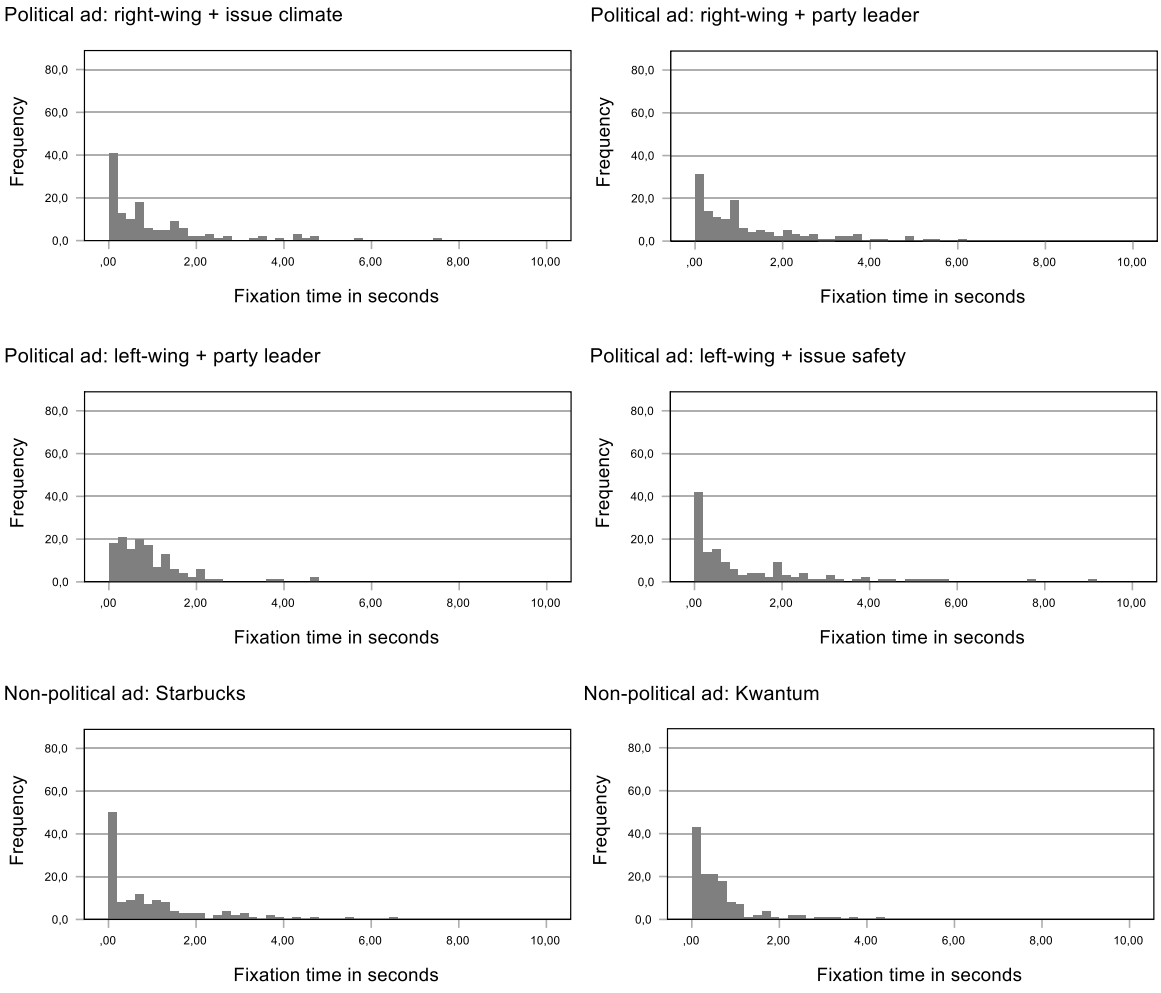


Figure 1. Distribution of total fixation duration per advertisement.

Hypothesis testing

We ran a series of linear regressions with vote intention and topic importance as independent variables and visual attention to the political advertisements as dependent variables. These results are presented in Table 2. In addition, we ran logistic regressions with vote intention and topic importance as independent variables and recall and recognition of the four political advertisements as dependent variables. The results for recall are presented in Table 3, and the results for recognition are presented in Table 4. Political interest, age, gender, education, the time participants spent reading the news articles, the order in which the stimuli were presented, and the number of days until the Dutch elections were added as control variables.[5] Note that for visual attention and recognition, we can analyze the results per political advertisement. However, for recall, we can only analyze the results with the advertisements combined per political party because we asked about the sender of the ad, rather than what was depicted

Table 2. Visual attention for the political advertisement.[6]

	1 Right-wing + party leader b (se)	2 Right-wing + climate b (se)	3 Left-wing + party leader b (se)	4 Left-wing + safety b (se)
Intercept	−0.80 (0.88)	−0.97 (0.99)	−.014 (0.57)	−0.96 (1.19)
Right-wing vote intention	0.02 (0.03)	−0.02 (0.03)		
Topic importance climate		0.18 (0.24)		
Right-wing * climate		0.12 (0.06)*		
Left-wing vote intention			0.04 (0.02)*	0.06 (0.04)
Topic importance safety				−0.03 (0.30)
Left-wing * safety				−0.10 (0.08)
R^2	.20	.27	.20	.21

Note: *$p < .05$; **$p < .01$; ***$p < .001$, political interest, age, gender, education, the time participants spent reading the news articles, the order in which the stimuli were presented, and the number of days until the Dutch elections were added as control variables.

Table 3. Recall of political advertisements.

Predictor	1 Right-wing ads b (se)	2 Left-wing ads b (se)
Right-wing vote intention	0.05 (0.06)	
Topic importance climate	−0.50 (0.47)	
Right-wing * climate	−0.06 (0.11)	
Left-wing vote intention		−0.03 (0.06)
Topic importance safety		−0.41 (0.44)
Left-wing * safety		−0.02 (0.12)
Nagelkerke R^2	.26	.21
−2 Log Likelihood	147.97	152.09
$X^2(12)$	28.88	22.41
p	.004	.033

Note: *$p < .05$; **$p < .01$; ***$p < .001$, political interest, age, gender, education, the time participants spent reading the news articles, the order in which the stimuli were presented, and the number of days until the Dutch elections were added as control variables.

Table 4. Recognition of political advertisements.

Predictor	1 Right-wing + party leader b (se)	2 Right-wing + climate b (se)	3 Left-wing + party leader b (se)	4 Left-wing + safety b (se)
Right-wing vote intention	0.02 (0.06)	−0.04 (0.06)		
Topic importance climate		−0.72 (0.46)		
Right-wing * climate		0.07 (0.11)		
Left-wing vote intention			0.03 (0.06)	0.08 (0.05)
Topic importance safety				0.69 (0.42)
Left-wing * safety				−0.09 (0.11)
Nagelkerke R^2	.27	.27	.15	.12
−2 Log Likelihood	151.13	156.17	146.95	174.03
X^2	(8) 29.74	(10) 30.08	(8) 15.36	(10) 12.52
p	<.001	<.001	.053	.252

Note: *$p < .05$; **$p < .01$; ***$p < .001$, political interest, age, gender, education, the time participants spent reading the news articles, the order in which the stimuli were presented, and the number of days until the Dutch elections were added as control variables.

in the ad. So, we only know whether participants recalled seeing a right-wing or left-wing ad, but not which of the two right-wing or left-wing advertisements they recall specifically.

To test H1, we examined whether right-wing voters allocated more visual attention to right-wing political advertisements compared to left-wing voters and vice versa. Results (see Table 2) reveal a significant effect of left-wing vote intention on visual attention toward the left-wing ad depicting the party leader (see Figure A1.2 in Appendix A), $b = 0.04$, $se = 0.02$, $\beta = 0.20$, $p = .033$. This means that left-wing vote intention was a significant driver of visual attention toward the left-wing ad depicting the party leader. Vote intention did not have a significant effect on the amount of visual attention participants allocated to the other three political advertisements. Hence, H1a was partly supported.

Next, to examine whether vote intention positively affects ad memory (i.e., recall and recognition), we ran two logistic regression analyses. Table 3 shows that there is no significant effect of vote intention on recall, meaning that right-wing voters did not recall right-wing ads more, and left-wing voters did not recall left-wing ads more. Table 4 shows that there is also no significant effect of vote intention on recognition, meaning that right-wing voters did not recognize right-wing ads better, and left-wing voters did not recognize left-wing ads better. Thus, H1b was rejected.

To test H2, we examined whether participants who consider climate an important topic allocated more visual attention to a political ad about climate, and if participants who consider safety an important topic allocated more visual attention to a political ad about safety. To test H2a, we ran linear regressions with topic importance as the independent variable and total fixation time of the political ads depicting political topics as the dependent variable. As shown in Table 2, topic importance did not have a significant effect on visual attention toward political advertisements. As such, H2a was rejected. To examine whether topic importance has a positive effect on ad memory (i.e., recall and recognition), we ran two logistic regression analyses. Table 3 shows that there is no significant effect of topic importance on recall, meaning that participants who found climate important did not recall climate advertisements more, and voters who found safety important did not recall safety advertisements more. Table 4 shows that topic importance does not significantly affect the recognition of political advertisements. Taken together, H2b was rejected.

To answer our research question, we examined whether there is an interaction effect between participants' vote intention and topic importance on visual attention, recall, and recognition of political advertisements. As shown in Table 2, we find a significant interaction effect of right-wing vote intention and the topic climate, $b = 0.12$, $se = 0.06$, $\beta = 0.50$, $p = .040$. Hence, right-wing participants who find climate an important issue allocate significantly more visual attention to the right-wing ad depicting the issue of climate (see Figure A1.3 in Appendix A). However, we did not find a similar effect for left-wing voters and the topic safety. Tables 3 and 4 show that no significant interaction effects on the recall or recognition of political ads. Thus, we find mixed support for the idea that political ads that match both participants' vote intention, as well as interest in a political topic, increase the amount of visual attention people allocate to the political ads, and in turn, their ability to remember political ads.

Discussion

This study examined whether political advertisements that align with individuals' political preferences draw more attention and are remembered better while performing a goal-directed task. As hypothesized, we found that left-wing vote intention significantly predicted visual attention toward the left-wing ad depicting the party leader (see Figure A1.3 in Appendix A). Likewise, stronger right-wing vote intention, in combination with greater perceived importance of the topic climate, significantly increased visual attention to the right-wing ad on that topic (see Figure A1.2 in Appendix A). These effects suggest that, under certain conditions, targeted political advertisements can indeed attract more visual attention, particularly when the political message aligns with voters' existing motivations or concerns.

At the same time, the broader pattern of results does not reveal consistent evidence of selective

exposure or biased processing. Overall, voters did not attend more to congruent than to incongruent political ads, nor did they recall or recognize them better. This indicates that while political targeting may occasionally succeed in capturing attention, its effects appear contingent on specific combinations of message content, voter orientation, and issue salience rather than being a generalizable phenomenon. This is an interesting and crucial finding, as it is often assumed in the popular debate that targeted political ads have a powerful impact on voters because they supposedly draw more attention and are remembered better. This study does not provide strong evidence for these firm convictions.

Our study shows that when reading news online, people do fixate on advertisements, but only briefly (see Figure 1 and Table 1). Because the majority of participants in our sample fixated on the political advertisements at least once, the lack of significant findings cannot be attributed to banner blindness. If banner blindness had occurred, most participants would not have fixated on the political advertisements at all. Our findings, therefore, suggest that while political advertisements attract initial visual attention, they fail to sustain it. This is key for political targeting literature as it challenges the often-made assumption that targeted political advertising has enduring persuasive effects. Instead, our results indicate that people retain considerable agency over which information they attend to and process, contrary to portrayals of voters as being passively steered by advertisers and online platforms in the current era of misinformation and generative AI. It should be noted, however, that some news articles may have been particularly engaging to certain participants, which could have suppressed attention to the accompanying political advertisement.

Moreover, these results add nuance to the normative debate on (micro)targeting as regulators have proposed stricter rules (Woollacott, 2023). Following the adoption of the transparency and targeting of political advertising regulation (TTPA) (European Union, 2024), Meta has decided to suspend the possibility of political advertising in European countries on their platforms (Kwai, 2025). This means that political advertising on Facebook and Instagram no longer occurs. It is, however, too early to determine whether political advertisements will truly disappear from Meta platforms or whether the rules can be circumvented. Studying the effects of political advertising on alternative platforms, such as news websites, has therefore become even more pressing, as Meta's decision may lead to an increase in political advertising on such platforms in the near future. One might question whether such measures are warranted if politically congruent advertisements do not attract greater visual attention or are not better remembered than incongruent ones. In this sense, targeted political advertisements can be seen as another form of information, with voters ultimately retaining control over what they engage with.

Interestingly, the results reported in this study partly contradict earlier studies by Schmuck et al. (2020) and Marquart et al. (2016), as both studies found that political preference significantly influences visual attention for political posters. However, in both studies, participants had to look at pairs of political posters (i.e., a left-wing poster accompanied by a right-wing poster or a neutral poster). When confronted with two political posters side by side, participants selectively exposed themselves to the poster that was closest to them. In our study, the stimulus material was presented next to a news article to create an ecologically valid scenario, as it is closer to ad exposure in everyday life. Based on our findings, we may conclude that people expose themselves selectively to congruent content primarily when choosing between congruent and incongruent content. When they encounter incongruent information on its own, people do not seem to avoid that information. This finding has, therefore, great implications for selective exposure literature in particular.

An alternative explanation for our findings could lie in the Dutch political landscape. During the 2023 elections, voters could vote for 26 different parties. Some of those parties are closely aligned. Although the parties used in this study differ from each other, neither is an extreme or radical political party. If we had chosen an extreme left party and a radical right party, selective exposure patterns might have been more clearly present, because the difference between the two parties is bigger. In addition, the Netherlands has rather low

levels of political polarization (Reiljan, 2020). If this study were conducted in a country with high levels of political polarization, such as the United States, we might find different patterns of selective exposure.

Our study is, of course, not without limitations. Although this study has provided a set of findings to understand political targeting in the Netherlands, some shortcomings should be noted. First, this study was part of a larger project that collected data across two different social media platforms with distinct goals. In both studies, participants were exposed to posts and ads. This has potentially primed them to pay more attention to the political advertisements within this study. Second, this study was conducted shortly before the 2023 Dutch general elections. Although we controlled for the number of days before the elections, it could still be the case that participants were primed toward political ads. Third, our choice for the parties VVD and GroenLinks-PvdA might represent a less common exposure scenario than selecting more ideologically similar parties. However, this choice was deliberate to maximize the likelihood of detecting differences in visual attention between distinct political parties. Choosing this scenario may somewhat limit the external validity of our findings. Moreover, choosing to depict issues with parties that do not own those issues might actually limit visual attention to those advertisements, as voters may discount messages from parties not strongly associated with that issue. Finally, knowing they were in an eye-tracking study could have altered their behavior. Hence, there is a possibility that participants looked longer at all of the advertisements because they expected to get questions about them. Consequently, the visual attention given to advertisements in this study may be overestimated compared to real-life behavior. However, despite the potential priming and increased attention, many participants did not fixate or only briefly fixated on the ads (see Figure 1), mitigating these limitations.

Taken together, we conclude that understanding the impact of political targeting is both more complicated and nuanced than previously thought. In addition, the potential persuasive advantage of political targeting does not lie in the first stages of message processing because people expose themselves to congruent and incongruent political ads equally. Contrary to common claims that alignment alone guarantees heightened attention or memory, our results point to a more nuanced picture in which targeting effects depend on contextual and motivational factors shaping voters' engagement with political messages.

Future studies should examine whether political targeting affects later stages of message processing by investigating the temporal effects and exploring the specific conditions under which targeted political advertisements attract more visual attention. In addition, eye-tracking studies examining the effect of political advertising should better mimic real-life behavior by enabling participants to scroll independently through a social media feed and by measuring behavioral intent. By doing so, the effects of political targeting and the extent to which people expose themselves to targeted political advertising can be better assessed. As our study did not allow participants to interact with the advertisements, and the ads did not include a call to action, future research should use more interactive environments to examine how political advertising influences actual behavior, such as clicking on an ad to obtain more information about the party. Lastly, future research should delve into potential country differences based on the levels of political polarization and the role of issue ownership shaping visual attention.

Notes

1. Anonymous link: https://osf.io/f7xnd/overview?view_only=1192ec3dcda0408a9feb93e8f75017b9.
2. The effect size used in this study resembles the effect size used in the study of Schmuck et al. (2016).
3. GroenLinks-PvdA and PVV had the same number of seats in the House of Representatives at the time of the experiment. Since GroenLinks-PvdA and VVD differ more from each other compared to PVV and VVD, we chose GroenLinks-PvdA for our experiment.
4. Topic importance was recoded into 1, 2, 3, 4, 5, 6 being important and 7 being very important because there was too little variation in the data.
5. The full regression tables including control variables can be found in Appendix D, E, and F.
6. For the political ads depicting the party leader we only add vote intention and the control variables to the model. The political ads about a topic also depict

a party logo (see Appendix A). Hence, for the political ads depicting a topic we add vote choice, topic importance, the control variables, and the moderation to the model. The fit statistics reported here are those of the complete model. All fit statistics can be viewed in Appendix D.

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Data availability statement

The data that support the findings of this study are available at https://doi.org/10.17605/OSF.IO/F7XND.

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